

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	Total Marks:	20 (Answer ANY ONE)
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
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Section / Batch:	B.Tech / 2024	Date:	07/08/2026

Code of Conduct

I S. Dharmika/192421094 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

10. Compare FDMA (30 kHz/channel) and TDMA (200 kHz/carrier, 8 slots) for the same 12 MHz spectrum allocation. Derive the total channel capacity for each scheme, showing full working for both. Present the resulting capacity numbers side by side, and justify the spectral-efficiency gap between the two approaches explaining specifically why TDMA achieves more channels per MHz despite requiring wider individual carrier bandwidth.



ITA0301 – Mobile Computing CO3-AT2 Technical Presentation

FDMA vs TDMA: Which utilizes the available spectrum more efficiently?

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Compare FDMA (30 kHz/channel) and TDMA (200 kHz/carrier, 8 slots) for the same 12 MHz spectrum allocation. Derive the total channel capacity for each scheme, showing full working for both. Present the resulting capacity numbers side by side, and justify the spectral-efficiency gap between the two approaches — explaining specifically why TDMA achieves more channels per MHz despite requiring wider individual carrier bandwidth.

Comparison of FDMA and TDMA for a 12 MHz Spectrum Allocation

Given:

- Total Spectrum = 12 MHz = 12,000 kHz

FDMA

- Channel bandwidth = 30 kHz/channel

TDMA

- Carrier bandwidth = 200 kHz/carrier
- 8 time slots per carrier



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• **FDMA Capacity Calculation**

Formula

- Number of Channels = Channel Bandwidth / Total Spectrum

Calculation

- $12000 / 300 = 400$ channels

FDMA Capacity

- **Total Capacity = 400 simultaneous users (channels)**

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TDMA Capacity Calculation

Number of Carriers

- $12000 / 200 = 60$ carriers

Channels per Carrier

- Each carrier has **8 time slots**
- $60 \times 8 = 480$

TDMA Capacity

- **Total Capacity = 480 simultaneous users (channels)**



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Spectral Efficiency

FDMA

- $400 / 12 = 33.33$
- 33.33 channels/MHz

TDMA

Why is TDMA More Spectrally Efficient?

Although TDMA uses a wider carrier bandwidth (200 kHz) compared to FDMA's 30 kHz channel, it achieves more channels per MHz because:

- FDMA: Each 30 kHz frequency is dedicated to one user continuously.
- TDMA: One 200 kHz carrier is shared by 8 users using different time slots.
- Users transmit one after another, not simultaneously on different frequencies.
- Frequency resources are reused over time, increasing the number of users without requiring additional spectrum.
- Fewer guard bands are needed compared to assigning many separate FDMA channels, improving spectrum utilization.



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CONCLUSION

Difference:

$480 - 400 = 80$ additional channels

$80 / 400 \times 100 = 20\%$

Final Result:

TDMA provides 480 channels, which is 80 more channels (20% higher capacity) than FDMA in the same 12 MHz spectrum.

